

Large Language Models

How ChatGPT Predicts Words

Grade 4 Technology & AI Literacy Lesson

Discover how AI programs like ChatGPT learn language and predict the next word!

1. What Are Large Language Models?

Large Language Models (LLMs) are big computer programs that learn how to understand and use words. They read millions of books, stories, and websites to learn how language works!

★ *Imagine a giant brain that reads millions of stories and learns to talk just like you and me!*

2. How Do Computers Understand Words?

■ Recognition

Computers look for patterns and find similarities between words — just like spotting that 'cat' and 'bat' rhyme in '-at'.

■ Breaking Words Apart

Computers split words into smaller pieces called tokens to better understand their meaning.

3. Meet ChatGPT!

ChatGPT is a friendly AI helper that talks and writes like a person! It was created by a company called OpenAI and is one of the most well-known Large Language Models in the world.

★ *ChatGPT = Your friendly AI writing and talking helper!*

4. How Does ChatGPT Learn?

■ Reading

ChatGPT learns by reading millions of sentences from books, websites, stories, and more — like reading the biggest library in the world!

■ Patterns

It remembers how words and sentences usually go together, like knowing 'good morning' comes before 'everyone!'

■ Improving

The more ChatGPT practises, the better it gets at predicting words — just like how you improve at reading every day!

5. Predicting the Next Word

It's like playing 'What comes next?' in a story. ChatGPT guesses the best word to continue based on all the patterns it learned!

■ *"The dog likes to ____" → play, bark, sleep*

■ *"I like to eat ____" → apples, pizza, ice cream*

■ *"My favourite fruit is a juicy ____" → mango, apple, banana*

6. What AI Can and Cannot Do

■ What AI Can Do

Answer questions quickly • Help write stories • Translate languages • Find information from many sources. It's like a super-fast reading helper!

■ What AI Cannot Do

Understand feelings or emotions • Know what is truly right or wrong • It only makes guesses based on patterns it has learned.

7. Using AI Responsibly

■ Check with Adults

Always ask a parent or teacher to help verify what AI tells you.

■ Keep Personal Info Safe

Never share your name, address, school, or photos with AI chatbots.

■ Use AI to Help, Not Cheat

AI is a helper for learning — use it to understand, then do your own work!

8. Key Takeaways

- ✓ An LLM is a big computer brain that learns words from millions of texts.
- ✓ ChatGPT predicts the next word by finding patterns it learned during training.
- ✓ AI can help but it needs adult guidance — it doesn't truly understand the world.
- ✓ Always use AI responsibly and never share personal information.
- ✓ You can be part of the exciting future of AI!

9. Discussion Questions

1. Have you ever used ChatGPT or another AI helper? What did you ask it?
2. How is ChatGPT similar to how you learned to read and write?
3. Why is it important to check AI answers with a trusted adult?
4. What is one thing you would like to use AI to help you with in school?

Any questions about ChatGPT and AI? Let's chat! ■